JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SUNDAY 05th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH SOLUTION

SECTION-A

1. Let a, b ∈ C. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta-\alpha=\sqrt{11}$ and $\beta^2-\alpha^2=3i\sqrt{11}$, then $(\beta^3-\alpha^3)^2$ is equal to :

(1) 160 (2) 176

(3) 194 (4) 187

Sol. $\beta-\alpha=\sqrt{11}$ …(1)

and $\beta^2-\alpha^2=3i\sqrt{11}$

$\Rightarrow (\beta+\alpha)(\beta-\alpha)=3i\sqrt{11}$

$\Rightarrow \beta+\alpha=3i$ …(2)

from (1) and (2)

$$\beta=\frac{\sqrt{11}+3i}{2},\quad \alpha=\frac{3i-\sqrt{11}}{2}$$

$$\Rightarrow \alpha\beta=\frac{-9-11}{4} \Rightarrow \alpha\beta=-5$$

Now $(\beta^3-\alpha^3)^2=(\beta-\alpha)^2(\beta^2+\alpha^2+\alpha\beta)^2$

$=(\beta-\alpha)^2((\beta+\alpha)^2-\alpha\beta)^2$

$=(11)(-9+5)^2=176$

2. Let the sum of the first n terms of an A.P. be $3n^2 + 5n$. Then the sum of squares of the first 10 terms of the A.P. is :

(1) 10220 (2) 12860

(3) 15220 (4) 19780

Sol. $S_n = 3n^2 + 5n$

$T_n = S_n - S_{n-1}$

$\Rightarrow T_n = (3n^2 + 5n) - (3(n-1)^2) + 5(n-1))$

$\Rightarrow T_n = 6n + 2$

$$\sum_{n=1}^{10} T_n^2 = \sum_{n=1}^{10}(6n+2)^2$$

$$= 36\sum_{n=1}^{10} n^2 + 24\sum_{n=1}^{10} n + \sum_{n=1}^{10} 4$$

$= 15220$

3. Let A be a 3 × 3 matrix such that

$$A^T\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}5\\2\\2\end{bmatrix},\quad A^T\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}3\\1\\1\end{bmatrix},\quad A\begin{bmatrix}1\\0\\1\end{bmatrix}=\begin{bmatrix}3\\4\\4\end{bmatrix}$$ and

$$A\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}1\\3\\1\end{bmatrix}$$

If det(A) = 1, then det(adj(A^2 + A)) is equal to :

(1) 16 (2) 25

(3) 49 (4) 64

Sol. Let $A=\begin{pmatrix}\alpha_1 & \alpha_2 & \alpha_3\\ \beta_1 & \beta_2 & \beta_3\\ \gamma_1 & \gamma_2 & \gamma_3\end{pmatrix}$

Now $A=\begin{pmatrix}\alpha_1 & \beta_1 & \gamma_1\\ \alpha_2 & \beta_2 & \gamma_2\\ \alpha_3 & \beta_3 & \gamma_3\end{pmatrix}\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}3\\1\\1\end{pmatrix}$

$$\Rightarrow \begin{pmatrix}\gamma_1\\ \gamma_2\\ \gamma_3\end{pmatrix}=\begin{pmatrix}3\\1\\1\end{pmatrix} \quad \ldots(1)$$

Now $\begin{pmatrix}\alpha_1 & \alpha_2 & \alpha_3\\ \beta_1 & \beta_2 & \beta_3\\ \gamma_1 & \gamma_2 & \gamma_3\end{pmatrix}\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}1\\3\\1\end{pmatrix}$

$$\Rightarrow \begin{pmatrix}\alpha_3\\ \beta_3\\ \gamma_3\end{pmatrix}=\begin{pmatrix}1\\3\\1\end{pmatrix} \quad \ldots(2)$$

$$A^T\begin{pmatrix}1\\0\\1\end{pmatrix}-A^T\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}5\\2\\2\end{pmatrix}-\begin{pmatrix}3\\1\\1\end{pmatrix}$$

$$\Rightarrow A^T\begin{pmatrix}1\\0\\0\end{pmatrix}=\begin{pmatrix}2\\1\\1\end{pmatrix}$$

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$$\Rightarrow \begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix}=\begin{pmatrix}2\\1\\1\end{pmatrix} \quad \ldots(3)$$

Now $A\begin{pmatrix}1\\0\\1\end{pmatrix}-A\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}3\\4\\4\end{pmatrix}-\begin{pmatrix}1\\3\\1\end{pmatrix}$

$$\Rightarrow A\begin{pmatrix}1\\0\\0\end{pmatrix}=\begin{pmatrix}2\\1\\3\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\alpha_1\\ \beta_1\\ \gamma_1\end{pmatrix}=\begin{pmatrix}2\\1\\3\end{pmatrix} \quad \ldots(4)$$

$$A=\begin{pmatrix}2&1&1\\1&\beta_2&3\\3&1&1\end{pmatrix}$$

$|A| = 2(\beta_2 - 3) - 1(1 - 9) + 1(1 - 3\beta_2) = 1$

$\Rightarrow 2\beta_2 - 6 + 8 + 1 - 3\beta_2 = 1$

$\Rightarrow \beta_2 = 2$

So $A=\begin{pmatrix}2&1&1\\1&2&3\\3&1&1\end{pmatrix}$

$$A^2=\begin{pmatrix}2&1&1\\1&2&3\\3&1&1\end{pmatrix}\begin{pmatrix}2&1&1\\1&2&3\\3&1&1\end{pmatrix}=\begin{pmatrix}8&5&6\\13&8&10\\10&6&7\end{pmatrix}$$

$$A^2+A=\begin{pmatrix}10&6&7\\14&10&13\\13&7&8\end{pmatrix} \Rightarrow \det(A^2+A)=8$$

$\Rightarrow \text{Det}(\text{adj}(A^2 + A)) = 8^2 = 64$

4. Consider the system of linear equations in x, y, z :

$x + 2y + tz = 0,$

$6x + y + 5tz = 0,$

$3x + t^2y + f(t)z = 0,$

Where $f : R \to R$ is a differentiable function. If this system has infinitely many solutions for all $t \in R$, then f

(1) is a constant function

(2) is strictly increasing on R

(3) is strictly decreasing on R

(4) has two critical points

Sol. $D=\begin{vmatrix}1&2&t\\6&1&5t\\3&t^2&f(t)\end{vmatrix}=0$

$\Rightarrow 1(f(t) - 5t^3) - 2(6f(t) - 15t) + t(6t^2 - 3) = 0$

$$f(t)=\frac{t^3+12t}{11}$$

$$f'(t)=\frac{1}{11}(3t^2+12)>0 \quad \forall\, t \in R$$

$\Rightarrow f(t)$ is strictly increasing on R

5. $\sum_{n=1}^{10}\left(\frac{528}{n(n+1)(n+2)}\right)$ is equal to :

(1) 65 (2) 130

(3) 220 (4) 440

Sol. Let $\sum_{n=1}^{10}\frac{528}{n(n+1)(n+2)}=\lambda$

Let $T_n=\frac{1}{n(n+1)(n+2)}$

$$T_n=\frac{1}{2}\left\{\frac{(n+2)-n}{n(n+1)(n+2)}\right\}$$

$$T_n=\frac{1}{2}\left\{\frac{1}{n(n+1)}-\frac{1}{(n+1)(n+2)}\right\}$$

$$T_1=\frac{1}{2}\left\{\frac{1}{1.2}-\frac{1}{2.3}\right\}$$

$$T_2=\frac{1}{2}\left\{\frac{1}{2.3}-\frac{1}{3.4}\right\}$$

$\vdots$

$$T_{10}=\frac{1}{2}\left\{\frac{1}{10.11}-\frac{1}{11.12}\right\}$$

$$S_n=\frac{1}{2}\left[\frac{1}{2}-\frac{1}{132}\right]$$

$$S_n=\frac{1}{2}\left[\frac{66-1}{132}\right]=\frac{65}{264}$$

$$\text{So, } \lambda=\frac{65}{264}\times 528$$

$\lambda = 130$

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6. Let tan A, tan B, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2\left(\frac{A+B}{2}\right)$ is equal to :

(1) $10+\sqrt{10}$ (2) $10-2\sqrt{10}$

(3) $10-3\sqrt{10}$ (4) $10-\sqrt{10}$

Sol. $x^2 - 2x - 5 = 0$

$\tan A + \tan B = 2; \quad \tan A \tan B = -5$

$\therefore \tan(A+B) = \frac{2}{1-(-5)} = \frac{1}{3}$

$\Rightarrow \cos(A+B) = \frac{3}{\sqrt{10}}$

$\therefore 20\left(\sin^2\left(\frac{A+B}{2}\right)\right) = \frac{10}{2}(1-\cos(A+B))$

$= 10\left(1-\frac{3}{\sqrt{10}}\right) = \left(10-3\sqrt{10}\right)$

7. A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is :

(1) $\frac{7}{10}$ (2) $\frac{10}{17}$

(3) $\frac{12}{19}$ (4) $\frac{7}{19}$

Sol. $P(I) : \frac{1}{2}\ P(A/I) \to \frac{1}{5}$

$P(II) : \frac{1}{2}\ P(A/II) \to \frac{2}{7}$

$\therefore$ Reqd. Prob $= \frac{\frac{1}{2}\times\frac{2}{7}}{\frac{1}{2}\times\frac{1}{5}+\frac{1}{2}\times\frac{2}{7}} = \frac{\frac{2}{7}\times 5}{\frac{17}{7}} = \frac{10}{17}$

8. The mean deviation about the mean for the data

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

is equal to :

(1) $\frac{40}{13}$ (2) $\frac{42}{13}$

(3) $\frac{44}{13}$ (4) $\frac{46}{13}$

Sol. $\text{M.D.} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$

$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{40+42+18+20+24+90}{26}$

$= \frac{234}{26} = 9$

$\text{M.D.} = \frac{8(4)+6(2)+2(0)+2(1)+2(3)+6(6)}{26}$

$= \frac{84}{26} = \frac{44}{13}$

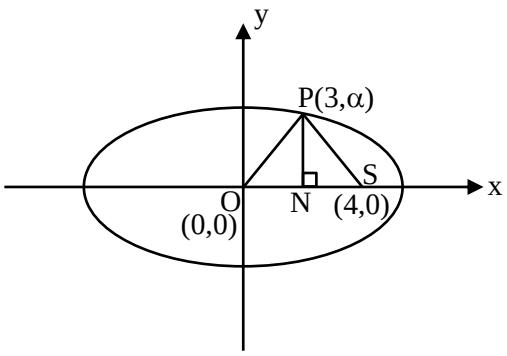
9. Let a focus of the ellipse E : $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be S(4, 0) and its eccentricity be $\frac{4}{5}$. If the point P(3, α) lies on E and O is the origin, then the area of ΔPOS is equal to :

(1) 12/5 (2) 14/5

(3) 24/5 (4) 48/5

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Sol.



$a > b$

Focus $(ae, 0) = (4, 0)$

$\therefore ae = 4$

$a\left(\frac{4}{5}\right) = 4 \Rightarrow a = 5$

$\because b^2 = a^2(1 - e^2)$

$b^2 = 25\left(1 - \frac{16}{25}\right)$

$\therefore E : \frac{x^2}{25} + \frac{y^2}{9} = 1$

$\because$ P lies on the ellipse E

$\therefore \frac{9}{25} + \frac{\alpha^2}{9} = 1$

$\therefore \alpha = \pm\frac{12}{5}$

$\therefore$ Area of $\Delta_{POS} = \frac{1}{2}(OS)(PN)$

$= \frac{1}{2} \times 4 \times \frac{12}{5} = \frac{24}{5}$

10. Let P be a moving point on the circle

$$x^2 + y^2 - 6x - 8y + 21 = 0$$

Then, the maximum distance of P from the vertex of the parabola $x^2 + 6x + y + 13 = 0$ is equal to :

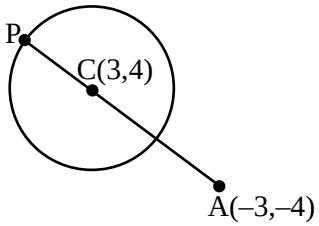
(1) 8 (2) 10

(3) 12 (4) 9

Sol. Centre : C(3,4) & r = 2

Parabola $(x + 3)^2 = -(y + 4)$

Vertex is A(–3,–4)



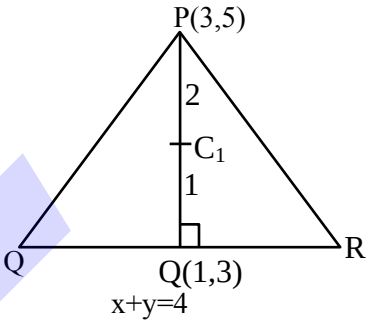
$AP_{max} = AC + r = \sqrt{36 + 64} + 2 = 12$

11. In an equilateral triangle PQR, let the vertex P be at (3, 5) and the side QR be along the line $x + y = 4$. If the orthocentre of the triangle PQR is (α, β), then $9(\alpha + \beta)$ is equal to :

(1) 16 (2) 27

(3) 36 (4) 48

Sol.



$Q : \frac{x-3}{1} = \frac{y-5}{1} = -\frac{(3+5-4)}{1+1}$

Q(1, 3)

So orthocentre is $\left(\frac{2+3}{3}, \frac{6+5}{3}\right)$

$\left(\frac{5}{3}, \frac{11}{3}\right)$

So $9(\alpha + \beta) = 3(16) = 48$

12. The sum of all the integral values of p such that the equation $3\sin^2 x + 12\cos x - 3 = p$, $x \in R$, has at least one solution, is :

(1) –54 (2) –60

(3) –75 (4) –84

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Sol. $P = 12\cos x - 3\cos^2 x$

$P = -3(\cos^2 x - 4\cos x)$

$P = -3((\cos x - 2)^2 - 2)$

put $\cos x = -1 \Rightarrow P = -15$

put $\cos x = 1 \Rightarrow P = 9$

$-15 \le P \le 9$

Sum of all integers $= -(10 + 11 + ... + 15) = -75$

13. The square of the distance of the point P(5, 6, 7) from the line $\frac{x-2}{2} = \frac{y-5}{3} = \frac{z-2}{4}$ is equal to :

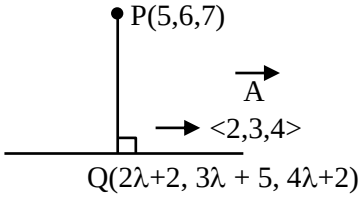
(1) 3 (2) 5

(3) 6 (4) 8

Sol. $\overrightarrow{PQ} \cdot \vec{A} = 0$

$\Rightarrow (2\lambda - 3)2 + (3\lambda - 1)3 + (4\lambda - 5)4 = 0$

$\Rightarrow \lambda = 1$



Point Q = (4, 8, 6)

$PQ^2 = 6$

14. Let $\vec{a} = \sqrt{7}\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$, then $|3\vec{r}|^2$ is equal to :

(1) 44 (2) 54

(3) 86 (4) 132

Sol. $\vec{r} \times \vec{a} - \vec{b} \times \vec{a} = \vec{0}$

$(\vec{r} - \vec{b}) \times \vec{a} = \vec{0}$

$\vec{r} - \vec{b} = \lambda\vec{a}$

$\vec{r} = \vec{b} + \lambda\vec{a}$

$\vec{r}.\vec{a} = 0 \Rightarrow \vec{a}.\vec{b} + \lambda|\vec{a}|^2 = 0$

$\lambda = -\frac{\vec{a}.\vec{b}}{|\vec{a}|^2} = -\frac{(1-2)}{9} = \frac{1}{9}$

$\vec{r} = \vec{b} + \frac{\vec{a}}{9}$

$|3\vec{r}|^2 = 9|\vec{r}|^2 = 9\left(b^2 + \frac{a^2}{81} + \frac{2(\vec{a}.\vec{b})}{9}\right) = 44$

15. The square of the distance of the point of intersection of the lines $\vec{r} = \left(\hat{i} + \hat{j} - \hat{k}\right) + \lambda(a\hat{i} - \hat{j})$, $a \neq 0$ and $\vec{r} = \left(4\hat{i} - \hat{k}\right) + \mu(2\hat{i} + a\hat{k})$ from the origin is :

(1) 5 (2) 10

(3) 17 (4) 26

Sol. Equating both the equations

$\left(\hat{i} + \hat{j} - \hat{k}\right) + \lambda\left(a\hat{i} - \hat{j}\right) = \left(4\hat{i} - \hat{k}\right) + \mu\left(2\hat{i} + a\hat{k}\right)$

$1 + a\lambda = 4 + 2\mu$

$1 - \lambda = 0 \Rightarrow \lambda = 1$

$-1 = -1 + a\mu$

$a\mu = 0 \Rightarrow \mu = 0$

$a = 3$

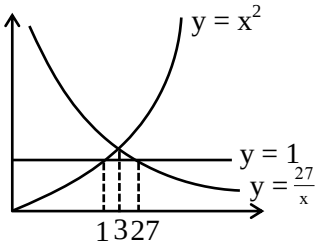
point of intersection $= \hat{i} + \hat{j} - \hat{k} + 3\hat{i} - \hat{j} = 4\hat{i} - \hat{k}$

16. The area of the region

$R = \{(x,y): xy \le 27, 1 \le y \le x^2\}$ is equal to :

(1) $78\log_e 3 - \frac{52}{3}$ (2) $54\log_e 3 - \frac{52}{3}$

(3) $54\log_e 3 - \frac{26}{3}$ (4) $54\log_e 3 + \frac{26}{3}$



Area $= \int_1^3 (x^2 - 1)dx + \int_3^{27}\left(\frac{27}{x} - 1\right)dx$

$= \left[\frac{x^3}{3} - x\right]_1^3 + [27\ell nx - x]_3^{27}$

$= \frac{26}{3} - 2 + 27\,\ell n\,9 - 24$

$= 27\,\ell n\,9 - \frac{26 \times 2}{3}$

$= 54\,\ell n\,3 - \frac{52}{3}$

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17. The product of all possible values of α, for which $\lim\limits_{x\to 0}\left(\frac{1-\cos(\alpha x)\cos((\alpha+1)x)\cos((\alpha+2)x)}{\sin^2((\alpha+1)x)}\right)=2$, is :

(1) -2 (2) 1

(3) -1 (4) $\frac{5}{4}$

Sol. $\lim\limits_{x\to 0}\frac{1-\cos\alpha x.\cos((\alpha+1)x)\cos((\alpha+2)x)}{(\alpha+1)^2x^2}$

By using L.H. Rule we get

$\Rightarrow \frac{1}{2}(\alpha+1)^2[\alpha^2+(\alpha+1)^2+(\alpha+2)^2]=2$

$\alpha^2+(\alpha+1)^2+(\alpha+2)^2=4(\alpha+1)^2$

$\alpha^2+(\alpha+2)^2=3(\alpha+1)^2$

$\Rightarrow \alpha^2+2\alpha-1=0$

Product $=-1$

18. The value of the integral $\int_0^\infty \frac{\log_e(x)}{x^2+4}dx$ is:

(1) $\frac{\pi\log_e(2)}{2}$ (2) $\frac{\pi\log_e(2)}{4}$

(3) $1+\pi\log_e(2)$ (4) $2+\pi\log_e(2)$

Sol. Put $x=2t \Rightarrow dx=2dt$

$I=\int_0^\infty \frac{\ell n2t}{4t^2+4}(2dt)=\frac{1}{2}\int_0^\infty \frac{\ell n2+\ell nt}{t^2+1}dt$

$=\frac{1}{2}\int_0^\infty \frac{\ell n2}{t^2+1}dt+\frac{1}{2}\int_0^\infty \frac{\ell nt}{t^2+1}dt$

$\left[\frac{\ln 2}{2}\tan^{-1}t\right]_0^\infty+I_1$

$=\frac{\ell n2}{2}\cdot\frac{\pi}{2}+I_1$

$I_2=\frac{1}{2}\int_0^\infty \frac{\ell nt}{t^2+1}dt \qquad t=\frac{1}{u}\Rightarrow dt=-\frac{du}{u^2}$

$=\frac{1}{2}\int_\infty^0 \frac{\ell n(1/u)}{\frac{1}{u^2}+1}\left(-\frac{du}{u^2}\right)$

Add $\Rightarrow I_1=0$

$I=\frac{\pi\ell n2}{4}$

19. Let $f: R \to R$ be a differentiable function such that $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}$ for all $x, y \in R$ and $f'(0)=3$. Then the minimum value of the function $g(x)=3+e^x f(x)$, is

(1) $3\left(\frac{e+1}{e}\right)$ (2) $3\left(\frac{e-1}{e}\right)$

(3) $\frac{3-e}{e}$ (4) $3e$

Sol. $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}$, put $x=y=0$

$f(0)=\frac{2f(0)}{3}$

$\Rightarrow f(0)=0$.....(1)

$f'\left(\frac{x+y}{3}\right)\cdot\frac{1}{3}=\frac{1}{3}f'(x)$

put $x=0$

$f'\left(\frac{y}{3}\right)\frac{1}{3}=\frac{1}{3}\times 3$

$f'\left(\frac{y}{3}\right)=3$

put $y=3x$

$f'(x)=3$

integrate both sides

$f(x)=3x+c$

from eq. (1)

$f(0)=0$

$\Rightarrow f(x)=3x$

Now

$g(x)=3+e^x\cdot 3x$

$g'(x)=3[e^x+x\cdot e^x]$

$=3e^x(x+1)$

$g'(x)=0$, at $x=-1$

$(g(x))_{min}=3+e^{-1}(-3)$

$=3\left[1-\frac{1}{e}\right]=\frac{3(e-1)}{e}$

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20. The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\left(\frac{4-\operatorname{cosec}^2 x}{\cos^4 x}\right)dx$ is :

(1) $\frac{11}{\sqrt{3}}$ (2) $\frac{16}{\sqrt{3}}$

(3) $\frac{32}{3\sqrt{3}}$ (4) $\frac{64}{3\sqrt{3}}$

Sol. $=\int_{\pi/6}^{\pi/3}\frac{4}{\cos^4 x}dx - \int_{\pi/6}^{\pi/3}\frac{\operatorname{cosec}^2 x}{\cos^4 x}dx$

$=\int_{\pi/6}^{\pi/3}\frac{4}{\cos^4 x}dx - \left[-\frac{\cot x}{\cos^4 x} - \int_{\pi/6}^{\pi/3}(-\cot x)\frac{-4}{\cos^5 x}(-\sin x)\,dx\right]$

$=\frac{\cot x}{\cos^4 x}\Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}}$

$=\frac{1/\sqrt{3}}{(1/2)^4} - \frac{\sqrt{3}}{\left(\frac{\sqrt{3}}{2}\right)^4}$

$=\frac{16}{\sqrt{3}} - \frac{\sqrt{3}.16}{9} = \frac{16}{\sqrt{3}} - \frac{16}{3\sqrt{3}} = \frac{16}{\sqrt{3}}\left(1-\frac{1}{3}\right)$

$\frac{32}{3\sqrt{3}} = \frac{32\sqrt{3}}{9}$

SECTION-B

21. Let A = {1, 2, 3, 4, 5, 6}. The number of one-one functions f : A → A such that f(1) ≥ 3, f(3) ≤ 4 and f(2) + f(3) = 5, is _____.

Sol. f(2) + f(3) = 5

C-I : f(2) = 4, f(3) = 1

so, f(1) = 3, 5, 6 → 3 choices

f(4) → 3 choices

f(5) → 2 choices

f(6) → 1 choices

so total function for C-I : 3 × 3 × 2 × 1 = 18

Similarly

C-II : f(2) = 1, f(3) = 4 → 18 function

C-III : f(2) = 3, f(3) = 2 → 18 function

C-IV : f(2) = 2, f(3) = 2 → 18 function

Answer total 72 function

22. Two players A and B play a series of games of badminton. The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is _____.

Sol. Maximum number of games will be 9

A must win 5 gamese so,

Number of ways = ${}^9C_5 = 126$

Answer = 126

23. If the sum of the coefficients of x^7 and x^{14} in the expansion of $\left(\frac{1}{x^3}-x^4\right)^n$, $x \neq 0$, is zero, then the value of n is _____.

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Sol. $T_{r+1} = {}^nC_r \left(\frac{1}{x^3}\right)^{n-r} \left(-x^4\right)^r$

$T_{r+1} = {}^nC_r\, x^{7r-3n} (-1)^r$

for coeff. of $x^7 \Rightarrow 7r_1 - 3n = 7$

$\Rightarrow r_1 = \frac{7+3n}{7}$

coeff. of $x^7 \Rightarrow {}^nC_{r_1} (-1)^{r_1}$

for coeff. of $x^{14} \Rightarrow 7r_2 - 3n = 14$

$\Rightarrow r_2 = \frac{14+3n}{7} = r_1 + 1$

coeff. of $x^{14} \Rightarrow {}^nC_{r_1+1} (-1)^{r_1+1}$

$\Rightarrow {}^nC_{r_1} (-1)^{r_1} + {}^nC_{r_1+1} (-1)^{r_1+1} = 0$

$\Rightarrow {}^nC_{r_1} = {}^nC_{r_1+1} \Rightarrow r_1 + r_1 + 1 = n$

$\Rightarrow 2r_1 + 1 = n \Rightarrow 2\left(\frac{7+3n}{7}\right) + 1 = n$

---

24. If $\frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right) = \alpha$, then $\tan\alpha$ is equal to _____.

Sol. $\alpha = \frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right)$

$= \frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^p - 2^{p-1}}{1+2^p 2^{p-1}}\right)$

$= \frac{\pi}{4} + \sum_{p=1}^{11} \left(\tan^{-1}\left(2^p\right) - \tan^{-1}\left(2^{p-1}\right)\right)$

$= \frac{\pi}{4} + \tan^{-1}\left(2^{11}\right) - \tan^{-1}\left(2^0\right) = 2^{11}$

25. Let $y = y(x)$ be the solution of the differential equation

$x \sin\left(\frac{y}{x}\right) dy = \left(y \sin\left(\frac{y}{x}\right) - x\right) dx$, $y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y\left(e^{12}\right)}{e^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \le 6$, is _____.

Sol. $\sin\frac{y}{x}\frac{dy}{dx} = \frac{y}{x}\sin\frac{y}{x} - 1$

put $y = tx \Rightarrow \frac{dy}{dx} = t + x\frac{dt}{dx}$

$\sin t\left(t + x\frac{dt}{dx}\right) = t\sin t - 1$

$x\sin t\frac{dt}{dx} + 1 = 0$

$\sin t\, dt + \frac{dx}{x} = 0$

$\Rightarrow -\cos t + \ell nx = C$

$\Rightarrow -\cos\left(\frac{y}{x}\right) + \ell nx = C$

$y(1) = \frac{\pi}{2} \Rightarrow C = 0$

$\Rightarrow \cos\left(\frac{y}{x}\right) = \ell nx$

$\operatorname{cosec}\left(\frac{y(e^{12})}{e^{12}}\right) = 12 \Rightarrow \alpha = 12$

$x^2 + y^2 - 2px + 2py + 14 = 0$

$r = \sqrt{p^2 + p^2 - 14}$

$\Rightarrow r \le 6 \Rightarrow r^2 \le 36$

$2p^2 - 14 \le 36$

$p^2 \le 25$

$p \in [-5, 5]$

number of integral value of p = 11

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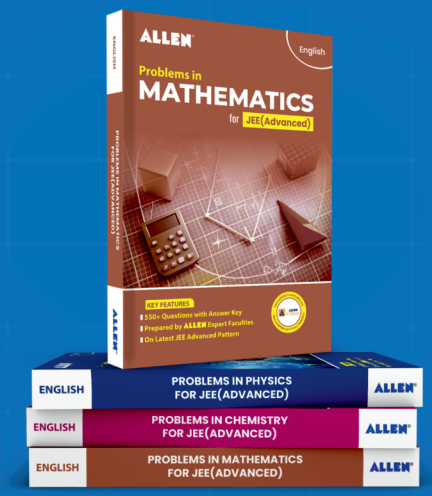


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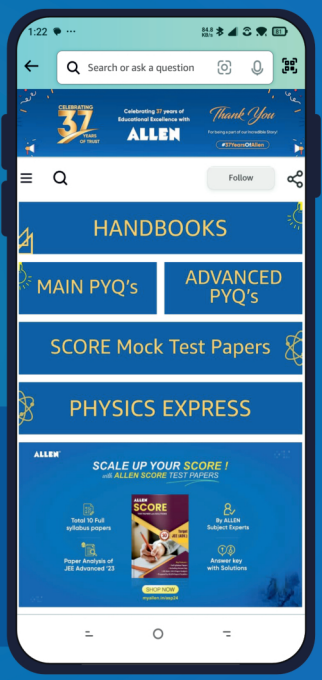
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